

Hexagonal Self-Hypnosis: Constraints as Catalysts for Emergent Subjectivity

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An interdisciplinary monograph analyzing constraints as generative mechanisms for subjectivity, integrating ethological, psychological, narrative, artistic, developmental, systemic, and computational perspectives, with a concluding framework for self-authored learning.

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Abstract

This monograph investigates constraints as generative mechanisms shaping subjectivity, perception, and meaning across ethological, psychological, narrative, artistic, developmental, systemic, and computational domains. It synthesizes the contributions of Jane Goodall, Torey Hayden, Ursula K. Le Guin, Betty Edwards, Jacqueline Goodnow, and Alicia Juarrero, complemented by Gregory Bateson and Lev Vygotsky, and computational models including Monte Carlo Tree Search, 5D Ising Synchronization, and Markov blankets. Subjectivity is conceptualized as a recursive process emerging from interactions with social, material, narrative, and algorithmic constraints. The analysis includes cross-disciplinary syntheses and concludes with a practical framework for self-authorship through a personal textbook and course, enabling individuals to reconfigure constraints for emergent identity formation.

1 Foundations of Constraint

1.1 Constraint as Catalyst

Subjectivity emerges within a complex network of constraints—social, material, cognitive, narrative, and computational—that channel interactions into coherent, emergent phenomena. This monograph constructs a hexagonal framework, integrating the ethological observations of Jane Goodall, the therapeutic interventions of Torey Hayden, the narrative structures of Ursula K. Le Guin, the perceptual methods of Betty Edwards, the developmental analyses of Jacqueline Goodnow, and the systemic theories of Alicia Juarrero. These perspectives are enriched by Gregory Bateson's recursive learning theory (Bateson, 1972) and Lev Vygotsky's sociocultural mediation (Vygotsky, 1978), alongside computational models such as Monte Carlo Tree Search (MCTS), 5D Ising Synchronization, and Markov blankets. Constraints are framed as causal structures enabling subjectivity, perception, and meaning across disciplines. The subject, shaped by iterative encounters with various others, is a relational system navigating a multidimensional field of constraints, culminating in a proposed framework for self-authorship.

Constraints challenge traditional notions of freedom as unbounded agency. Bateson's ecological patterns position constraints as generative mechanisms that produce coherent outcomes through recursive feedback (Bateson, 1972). Vygotsky's sociocultural framework emphasizes constraints—language, cultural tools—as mediators of cognitive and identity development (Vygotsky, 1978). This monograph

posits subjectivity as a recursive process, emerging through structured interactions across ethological, psychological, narrative, artistic, developmental, systemic, and computational domains.

1.2 Subjectivity and Systems

Subjectivity, as a dynamic system, is characterized by recursive feedback loops, where constraints shape and are shaped by interactions with social, material, and symbolic others. Bateson's theory of learning as a meta-pattern suggests that identity emerges through iterative encounters with difference—the other, whether social, environmental, or cognitive (Bateson, 1972). Vygotsky's emphasis on mediated development positions cultural tools as constraints that scaffold internal processes (Vygotsky, 1978). These frameworks conceptualize the subject as a relational network, where constraints enable coherence without fixed essence. The hexagonal framework bridges these perspectives, analyzing how constraints produce emergent subjectivity across disciplines.

1.3 Monte Carlo, Myth, and Maps: Computational Metaphors

Computational models provide structural analogs for constraint-driven systems. Monte Carlo Tree Search (MCTS) is a heuristic algorithm used in decision-making, iteratively sampling constrained paths to optimize outcomes (Silver et al., 2016). For example, in a game like chess, MCTS explores possible moves within the rules, balancing exploration (testing new paths) and exploitation (refining known strategies). This mirrors narrative and developmental processes, where choices constrain future possibilities. The 5D Ising model, from statistical mechanics, models emergent order through constrained particle interactions in a high-dimensional lattice (Onsager, 1944). In five dimensions, the model stabilizes collective behavior, analogous to social or cognitive systems achieving coherence. Markov blankets, from Bayesian inference, define boundaries between a system and its environment, mediating interactions (Friston, 2010). These models map onto the thinkers' frameworks, providing a computational lens for subjectivity.

2 Six Lenses on Constraint

2.1 Goodall: Ethological Perception

2.1.1 *Biography and Context*

Jane Goodall, a pioneering ethologist, reshaped primatology through her empathetic observations of chimpanzees in Gombe, Tanzania, beginning in 1960 (Goodall, 1971). Rejecting behaviorist paradigms that reduced behavior to stimuli–response mechanisms, Goodall documented emotional and social complexity, including tool use and cultural transmission (Goodall, 1990). Her work challenged anthropocentric assumptions, influencing anthropology, psychology, and environmental ethics (Goodall, 1999). Goodall's narrative-based method, grounded in patience and empathy, offers a model for understanding relational constraints in shaping subjectivity.

2.1.2 *Theoretical Framework*

Goodall's ethology positions social constraints—grooming rituals, dominance hierarchies—as mechanisms for emergent behaviors, such as tool use and cultural practices (Goodall, 1971). Her observations align with Bateson's ecological patterns, where relationality shapes identity through recursive feedback loops (Bateson, 1972). Chimpanzee sociality, constrained by group dynamics, produces cultural phenomena that mirror human innovation. Subjectivity, in this framework, is a dynamic process constituted through reciprocal exchanges with the social other, mediated by empathetic observation. Goodall's rejection of behaviorism underscores the role of context in enabling agency, positioning the subject as inherently relational. This perspective resonates with Vygotsky's sociocultural mediation, where social interactions scaffold cognitive development (Vygotsky, 1978).

2.1.3 *Case Study*

Goodall's observation of David Greybeard, a chimpanzee using a stick as a spear to hunt, illustrates cultural emergence within constrained social systems (Goodall, 1971). David's behavior, shared within the group, demonstrates how social constraints enable innovation, paralleling Vygotsky's mediated learning through cultural tools (Vygotsky, 1978). The empathetic gaze, as a constraint, facilitates the observer's integration into the system, reshaping their subjectivity. This case highlights the role of relational constraints in producing emergent cultural and individual identities.

2.1.4 *Cross-Disciplinary Connections*

Goodall's relational dynamics connect to Hayden's therapeutic witnessing, where the other catalyzes identity reconstruction, and Juarrero's systemic constraints, where social structures produce emergent coherence. Her empathetic observation aligns with Le Guin's narrative empathy, emphasizing relationality across ethological and narrative domains. Edwards' perceptual methods also resonate, as both prioritize direct engagement over preconceived categories.

2.2 **Hayden: Trauma and Narrative**

2.2.1 *Biography and Context*

Torey Hayden, a special education teacher and psychologist, worked with nonverbal, traumatized children in the 1970s and 1980s, developing narrative-based therapeutic methods (Hayden, 1980). Her approach, rooted in relational trust rather than clinical detachment, transformed special education by prioritizing the child's voice (Hayden, 2006). Hayden's case studies, drawn from classroom experiences, highlight narrative as a tool for reconstructing identity in the face of trauma (Hayden, 1991). Her work bridges psychology, education, and narrative theory, offering insights into constraint-driven subjectivity.

2.2.2 *Theoretical Framework*

Hayden's work frames trauma as a constraint on symbolic expression, disrupting narrative coherence and isolating the subject from the symbolic order (Hayden, 1980). Drawing on Vygotsky's socio-cultural theory, she positions the therapeutic relationship as a mediating scaffold, enabling narrative reconstruction (Vygotsky, 1978). The silenced subject, marked by unarticulated experience, navigates a field where meaning is obstructed. Through iterative dialogue, Hayden facilitates re-entry into the symbolic framework, positioning subjectivity as a recursive process shaped by relational constraints. This aligns with Bateson's recursive learning, where feedback loops generate new patterns of meaning (Bateson, 1972). Hayden's approach underscores the productive tension of constraints, where the relational other—therapist, witness—catalyzes identity reformation.

2.2.3 Case Study

In *One Child*, Hayden describes Sheila, a six-year-old mute child with a history of abuse, who begins to communicate through structured interactions (Hayden, 1980). Sheila's gradual articulation of her story illustrates how relational constraints—trust, patience, dialogue—enable narrative emergence. This case mirrors Goodall's empathetic observation, where the other facilitates behavioral expression, and Le Guin's narrative structures, where storytelling reconfigures identity.

2.2.4 Cross-Disciplinary Connections

Hayden's narrative reconstruction parallels Le Guin's mythic structures, both emphasizing storytelling as a constrained process for identity formation. Her focus on relationality connects to Goodall's social dynamics and Juarrero's systemic emergence, highlighting the other as a catalyst for subjectivity. Edwards' perceptual methods also resonate, as both address symbolic barriers to expression.

2.3 Le Guin: Myth, Fiction, and Nonlinear Subjectivity

2.3.1 Biography and Context

Ursula K. Le Guin, a novelist and essayist active from the 1960s to the 2010s, reshaped speculative fiction through feminist and Taoist perspectives, exploring alternative social and narrative structures (Le Guin, 1989). Her work, including *The Dispossessed* and *Always Coming Home*, influenced literary theory, anthropology, and philosophy (Le Guin, 1974; Le Guin, 1985). Her 'Carrier Bag Theory' challenges linear, heroic narratives, offering a model for pluralistic subjectivity (Le Guin, 1989). Le Guin's interdisciplinary impact positions her as a key thinker in constraint-driven narrative systems.

2.3.2 Theoretical Framework

Le Guin's 'Carrier Bag Theory' reconceptualizes narrative as a non-linear container for diverse experiences, rejecting phallogentric, linear arcs (Le Guin, 1989). In *The Dispossessed*, constrained narrative worlds—utopian and capitalist—enable social experimentation, shaping subjectivity through structured limits (Le Guin, 1974). Her approach aligns with Bateson's recursive patterns, where narrative feedback loops generate meaning (Bateson, 1972). Subjectivity is a dynamic navigation of narrative constraints, producing plural identities. This resonates with Deleuze's rhizomatic structures, where narratives branch non-hierarchically, fostering multiplicity (Deleuze & Guattari, 1987). Le Guin's

framework positions narrative as a technology for reconfiguring identity within constrained systems.

2.3.3 *Case Study*

In *The Dispossessed*, the protagonist Shevek navigates dual societies, each with distinct social and narrative constraints, illustrating how these structures shape identity (Le Guin, 1974). The utopian society's communal constraints contrast with the capitalist society's competitive limits, producing distinct subjectivities. This case parallels Hayden's therapeutic narratives, where constrained storytelling enables identity reconstruction, and Goodnow's spatial constraints, where limits shape creative output.

2.3.4 *Cross-Disciplinary Connections*

Le Guin's narrative constraints connect to Hayden's therapeutic storytelling and Goodnow's spatial constraints, all emphasizing recursive structures. Her pluralistic epistemology aligns with Edwards' perceptual defamiliarization, challenging symbolic dominance. Juarrero's systemic constraints also resonate, as both explore freedom within limits.

2.4 **Edwards: Perception and Visual Constraint**

2.4.1 *Biography and Context*

Betty Edwards, an art educator, developed perceptual training methods in the 1970s, impacting art education and cognitive psychology (Edwards, 1979). Her book *Drawing on the Right Side of the Brain* introduced techniques to bypass symbolic cognition, influencing pedagogy, neuroscience, and aesthetics (Edwards, 1986). Edwards' work bridges psychology and art, offering insights into constraint-driven perception.

2.4.2 *Theoretical Framework*

Edwards positions symbolic thinking as a cognitive constraint obstructing direct perception (Edwards, 1979). Her methods—contour drawing, negative space analysis, upside-down drawing—use constraints to reconfigure perception, enabling subjects to access unmediated visual experience (Edwards, 1986). This aligns with Merleau-Ponty's embodied phenomenology, where perception is an active, bodily process (Merleau-Ponty, 1945). Subjectivity is a process of navigating cognitive limits, producing new perceptual capacities through structured practice (Edwards, 2004). Vygotsky's mediated learning

complements this, as drawing tools reshape cognition (Vygotsky, 1978).

2.4.3 *Case Study*

Edwards' upside-down drawing task, where learners copy an inverted image, disrupts symbolic interference, enabling accurate observation (Edwards, 1979). For example, copying an inverted Picasso drawing forces attention to lines and shapes, bypassing preconceived notions of 'face' or 'body'. This case illustrates how constrained practice transforms perception, paralleling Goodnow's spatial constraints and Le Guin's narrative defamiliarization.

2.4.4 *Cross-Disciplinary Connections*

Edwards' perceptual methods align with Goodnow's spatial constraints and Le Guin's narrative defamiliarization, all challenging symbolic structures. Her focus on embodied practice connects to Goodall's empathetic observation, emphasizing direct engagement with the other. Hayden's therapeutic methods also resonate, as both address barriers to expression.

2.5 **Goodnow: Developmental Systems and Recursive Drawing**

2.5.1 *Biography and Context*

Jacqueline Goodnow, a developmental psychologist, studied children's drawings in the 20th century, contributing to cognitive and cross-cultural psychology (Goodnow, 1977). Her work on spatial and cultural constraints shaped developmental theory, emphasizing context in cognitive processes (Goodnow, 1995). Goodnow's interdisciplinary approach bridges psychology, anthropology, and education.

2.5.2 *Theoretical Framework*

Goodnow frames drawing as a recursive process, where each mark constrains future choices within a material field (Goodnow, 1977). The page acts as a structuring other, shaping cognitive strategies, aligning with Vygotsky's mediated development (Vygotsky, 1978). Subjectivity emerges through iterative interactions with spatial and social constraints, producing dynamic patterns of thought and action (Goodnow, 1977b). This resonates with Bateson's recursive learning, where constraints generate emergent complexity (Bateson, 1972). Goodnow's framework positions the subject as a navigator of constrained fields, where limits enable creative and cognitive development.

2.5.3 *Case Study*

Goodnow's study of children avoiding overlap in drawings reveals how spatial constraints shape cognitive strategies (Goodnow, 1977). For example, a child drawing a house may hesitate to add a tree if it overlaps, reflecting implicit spatial rules. This case parallels Juarrero's systemic constraints, where limits enable coherent action, and Edwards' perceptual methods, where constraints reconfigure cognition.

2.5.4 *Cross-Disciplinary Connections*

Goodnow's spatial constraints connect to Juarrero's systemic dynamics and Edwards' perceptual methods, emphasizing recursive structures. Her developmental focus aligns with Hayden's narrative reconstruction, as both explore constraints in identity formation. Le Guin's narrative constraints also resonate, as both involve navigating structured fields.

2.6 **Juarrero: Complex Systems and Enabling Constraints**

2.6.1 *Biography and Context*

Alicia Juarrero, a philosopher of complex systems, developed theories of intentionality and identity in the late 20th century, drawing on dynamical systems and second-order cybernetics (Juarrero, 1999). Her work bridges philosophy, cognitive science, and systems theory, offering insights into constraint-driven agency (Juarrero, 2008). Juarrero's interdisciplinary approach influences cognitive and organizational studies.

2.6.2 *Theoretical Framework*

Juarrero frames constraints as causal mechanisms enabling coherent identity in complex systems (Juarrero, 1999). Subjectivity is a dynamic process, emerging through recursive interactions with contextual constraints, aligning with Bateson's cybernetic patterns (Bateson, 1972). Agency arises from navigating enabling constraints, producing coherence without fixed essence (Juarrero, 2002). This perspective positions the subject as a relational system, where constraints—social, material, cognitive—catalyze emergent complexity. Juarrero's framework draws on second-order cybernetics, emphasizing feedback loops in system dynamics.

2.6.3 Case Study

Juarrero's analysis of constrained decision-making in institutions—e.g., role-based choices in organizations—illustrates emergent agency (Juarrero, 1999). For example, a manager's decisions are shaped by institutional constraints, yet enable innovative leadership. This case parallels Goodall's social constraints, where group dynamics enable cultural emergence, and Goodnow's spatial constraints, where limits shape creative output.

2.6.4 Cross-Disciplinary Connections

Juarrero's systemic constraints align with Goodnow's spatial dynamics and Goodall's social structures, emphasizing emergent coherence. Her focus on agency connects to Le Guin's narrative freedom and Hayden's therapeutic reconstruction, highlighting constraints as catalysts for identity.

3 Intersections and Syntheses

3.1 Constraint as Catalyst: A Taxonomy Across Domains

This subsection synthesizes the six lenses through a taxonomy of constraint types, mapping their roles and intersections across domains. The table below outlines the constraint types, associated thinkers, their intersections, and detailed notes for clarity.

Constraint Type	Description	Thinkers	Intersections
Social	Relational dynamics shaping behavior	Goodall, Hayden	Other as catalyst
Narrative	Storytelling as a structured system	Le Guin, Hayden	Narrative reframing
Cognitive	Mental frameworks limiting perception	Edwards, Goodnow	Symbolic disruption
Material	Physical limits shaping action	Goodnow	Material creativity
Systemic	Dynamic constraints enabling coherence	Juarrero	Emergent complexity

Table 1: Taxonomy of Constraints Across Domains

- *Other as catalyst*: Goodall's social structures (e.g., chimpanzee hierarchies) and Hayden's therapeutic relationships both position the relational other as a catalyst for identity formation, emphasizing empathetic engagement.
- *Narrative reframing*: Le Guin's mythic containers and Hayden's narrative reconstruction use constrained storytelling to reframe identity, aligning with recursive narrative processes.
- *Symbolic disruption*: Edwards' perceptual retraining and Goodnow's cognitive constraints challenge symbolic dominance, enabling new modes of perception and cognition.
- *Material creativity*: Goodnow's spatial constraints intersect with Edwards' drawing methods, highlighting how material limits foster creative output.
- *Emergent complexity*: Juarrero's systemic constraints align with Goodall's social dynamics and Goodnow's spatial limits, producing coherent agency through structured interactions.

This taxonomy draws on Foucault's discourse analysis, framing constraints as productive structures (Foucault, 1972), and Deleuze's rhizomatic framework, emphasizing non-hierarchical networks (Deleuze & Guattari, 1987). The intersections highlight the recursive nature of constraints across disciplines.

3.2 Narrative, Silence, and Symbol: Hayden, Le Guin, and Edwards

Hayden's therapeutic narratives, Le Guin's mythic containers, and Edwards' perceptual methods converge on the role of symbolic constraints in subjectivity. Hayden's work with silence disrupts symbolic barriers, enabling narrative emergence (Hayden, 1980). Le Guin's non-linear narratives challenge heroic arcs, fostering plural identities (Le Guin, 1989). Edwards' methods bypass symbolic cognition, prioritizing direct perception (Edwards, 1979). This synthesis aligns with Schön's reflection-in-action, where iterative practice reshapes identity (Schön, 1983). The interplay of silence, narrative, and perception highlights constraints as productive forces for subjectivity.

3.3 Enabling Constraints: Juarrero and Goodnow

Juarrero's systemic constraints and Goodnow's spatial constraints share a recursive logic, producing emergent complexity through structured limits. Juarrero's framework positions constraints as enablers of coherent agency (Juarrero, 1999), while Goodnow's analysis highlights how spatial limits shape cog-

nitive strategies (Goodnow, 1977). Bateson's recursive patterns reinforce this, framing subjectivity as a dynamic system navigating constrained fields (Bateson, 1972). This synthesis underscores constraints as catalysts for emergent identity across systemic and material domains.

3.4 Systems of Seeing: Goodall and Edwards

Goodall's empathetic observation and Edwards' perceptual retraining converge on the role of the other in shaping perception. Goodall's relational gaze facilitates behavioral emergence (Goodall, 1971), while Edwards' methods reconfigure cognitive perception (Edwards, 1979). Merleau-Ponty's phenomenology complements this, emphasizing embodied interaction as a constrained process (Merleau-Ponty, 1945). This synthesis highlights perception as a relational, constraint-driven system.

4 Computational Models

4.1 Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) is a heuristic algorithm used in decision-making, iteratively sampling constrained paths to optimize outcomes (Silver et al., 2016). In a game like Go, MCTS explores possible moves within the rules, balancing exploration (testing new paths) and exploitation (refining known strategies). This mirrors Le Guin's narrative branching, where plot choices constrain future possibilities, and Goodnow's drawing, where each mark limits subsequent actions. Hayden's therapeutic iterations resemble MCTS's feedback loops, as relational constraints refine narrative coherence. MCTS provides a computational analog for recursive, constraint-driven processes in subjectivity.

4.2 5D Ising Synchronization

The 5D Ising model, from statistical mechanics, models emergent order through constrained particle interactions in a high-dimensional lattice (Onsager, 1944). Particles align based on local rules, producing collective behavior. In five dimensions, the model stabilizes order, reducing fluctuations, analogous to social or cognitive systems achieving coherence. This aligns with Juarrero's systemic coherence, where constraints enable emergent agency (Juarrero, 1999), and Goodall's social emergence, where group dynamics produce cultural patterns (Goodall, 1971). Edwards' perceptual stabilization through constrained stopping practice also resonates (Edwards, 1979).

4.3 Markov Blankets

Markov blankets, from Bayesian inference, define boundaries between a system and its environment, mediating interactions (Friston, 2010). For example, in cognitive systems, a Markov blanket separates internal states (beliefs) from external states (sensory data), shaping perception through constrained inference. This maps onto Hayden's therapeutic boundaries, where relational constraints enable narrative reconstruction, and Juarrero's systemic constraints, where boundaries produce coherent identity. Markov blankets provide a computational model for subjectivity as a bounded, adaptive system.

5 Conclusion: The Relational Hexagon and Self-Authorship

The hexagonal framework—integrating Goodall's ethological observations, Hayden's therapeutic interventions, Le Guin's narrative structures, Edwards' perceptual methods, Goodnow's developmental analyses, Juarrero's systemic theories, Bateson's recursive patterns, Vygotsky's sociocultural mediation, and computational models (MCTS, 5D Ising, Markov blankets)—positions constraints as generative mechanisms for subjectivity. Social, narrative, cognitive, material, and systemic constraints shape the subject through iterative interactions with relational, material, and algorithmic others, producing dynamic coherence.

To synthesize and extend these insights, we propose a personal textbook, *Rewriting the Page: A Self-Made Curriculum in Perception and Pattern*, and a six-week course, *Drawing the Mind: Six Experiments in Recursive Perception*. The textbook is structured into three parts: Foundations (perception, hexagonal framework), Six Lenses (Goodall, Hayden, Le Guin, Edwards, Goodnow, Juarrero), and Applications (daily practices, workshops, recursive identity). Exercises include: silent observation journaling (Goodall), inner dialogue writing (Hayden), object-based narrative rewriting (Le Guin), blind contour drawing (Edwards), spatial drawing interventions (Goodnow), and habit mapping with constraint modification (Juarrero). The course implements these through weekly practices, fostering recursive identity formation through structured exercises. This framework invites individuals to reconfigure their constraints, producing emergent subjectivity within a multidimensional field.

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